

SUMMARY

This video explains the foundations, and preparation strategy outlined in the introductory session of the GATE 2026 Discrete Mathematics Crash Course for CS and IT aspirants.

Key highlights:

Comprehensive coverage from fundamentals to advanced problem solving.

Condensed 15-day structured schedule.

Computer Science–specific interpretations (e.g., 0 included in Natural Numbers).

Emphasis on bijections for infinite set equivalence.

Strong focus on mock analysis and high-weightage topics.

I. INSTRUCTIONAL SCHEDULE AND COURSE STRUCTURE

The course follows a compressed yet systematic timeline to complete Discrete Mathematics before year-end.

Weekly Topic Breakdown

Date	Topic / Content
Dec 16	Introduction to Sets
Dec 17	Basic Types of Relations
Dec 18	Equivalence Relations and Partial Order Relations
Dec 19	Hasse Diagrams
Dec 20	Functions
Dec 21	Sunday Special: PYQs (Sets, Relations, Functions)
Dec 22	Group Theory
Dec 23–25	Graph Theory (3-day coverage)
Dec 26–27	Propositional and Predicate Logic
Dec 28	Sunday Special: Weekly PYQs
Dec 29	Recurrence Relations

Dec 30	Generating Functions
Jan 2–3	Basics of Permutations and Combinations

Session Timing

Sessions begin at 9:00 PM and continue until topic completion.
Duration may extend beyond midnight.

II. SET THEORY FUNDAMENTALS

2.1 Core Definitions

Set

An unordered collection of distinct elements.
Duplicates are not allowed.
Elements may be heterogeneous.

Roster Form

Explicit listing of elements.
Example: $\{1, 7, 9, 3, A\}$

Set-Builder Form

Defined by property.
Example: $\{x \mid 0 \leq x \leq 1, x \in \mathbb{R}\}$

2.2 Cardinality and Types of Sets

Cardinality ($|A|$)

Number of elements in a set.

Finite Set

Has a fixed number of elements.

Infinite Set

Has infinitely many elements.

Empty Set ($\emptyset$ or $\{\}$)

Contains no elements.
Cardinality = 0.

Important distinction:

$\{\} \rightarrow$ Empty set

$\{\emptyset\} \rightarrow$ Contains one element (the empty set)

Cardinality = 1

2.3 Natural Numbers in CS Context

For GATE CS/IT:

$$0 \in \mathbb{N}$$

This aligns with:

Array indexing
Memory addressing
Page table numbering

III. EQUALITY VS EQUIVALENCE OF SETS

3.1 Equivalent Sets

Finite Sets

Equivalent if:

$$|A| = |B|$$

Infinite Sets

Equivalent if and only if a bijective function exists between them.

Example

Multiples of 2 $\leftrightarrow$ Natural Numbers

Bijjective mapping: $f(x) = x/2$

Non-equivalence

Natural Numbers $\neq$ Real numbers in $(0,1)$

No bijection exists.

3.2 Equal Sets

$A = B$ if:

$$A \subseteq B \text{ and } B \subseteq A$$

Implication

Equal sets are equivalent.

Equivalent sets are not necessarily equal.

IV. SUBSETS, POWER SETS, AND SUPERSETS

4.1 Subsets

$$A \subseteq B$$

Every element of A belongs to B.

Universal truths:

$$A \subseteq A$$

$$\emptyset \subseteq A$$

Proper Subset

Subset excluding the set itself.

4.2 Power Set

Definition

$P(A)$ = set of all subsets of A

Cardinality

If $|A| = n$

$$|P(A)| = 2^n$$

Identity

$$C(n,0) + C(n,1) + \dots + C(n,n) = 2^n$$

4.3 Super Sets

If $A \subseteq B$

Then $B \supseteq A$

Number of Super Sets

Let universal set U have n elements.

Let A have m elements.

Number of super sets of A:

$$2^{n-m}$$

V. BIJECTIVE FUNCTIONS

A function is bijective if it is both injective and surjective.

Injective (One-to-One)

Distinct domain elements map to distinct codomain elements.

$$|\text{Domain}| \leq |\text{Codomain}|$$

Surjective (Onto)

Every codomain element has a pre-image.

$$|\text{Domain}| \geq |\text{Codomain}|$$

Bijjective

$$|\text{Domain}| = |\text{Codomain}|$$

VI. STRATEGIC GUIDANCE FOR GATE PREPARATION

6.1 Revision Strategy

If notes are complete:

Full subject revision in 2–3 days.

If revision neglected:

10–12 hours revision daily

3 hours practice daily

6.2 Mock Test Strategy

Minimum 15 full-length mock tests recommended.

Key principle:

Analysis > Quantity

After each mock:

Identify weak areas

Diagnose conceptual gaps

Improve time management

Revise before next mock

6.3 COA Topic Prioritization

High-weightage topics:

Cache Memory Organization

Pipelining

Direct Memory Access (DMA)

